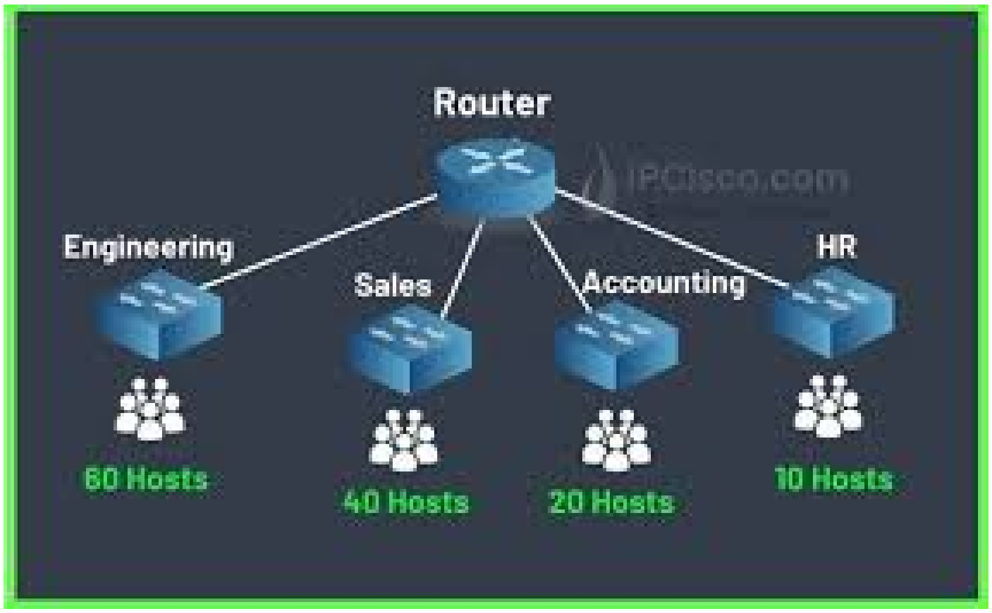
NETWORKING
VLSM Subnetting in Class-C

Subnetting using Variable Length Subnet Mask method is a very useful way when we have any one of the following requirements.

- **Variable Number of Hosts in each subnet**
  There can be requirements to divide a network into smaller networks having different numbers of hosts unlike FLSM where all subnets contain equal numbers of IPs.
- **To overcome IP addresses wastage**
  We can minimize IPs wastage while sub-netting through VLSM technique.

Scenario: Let's have an IP **192.168.3.0/24** we have to create subnets.



As shown in the diagram we have to create four different subnets with different numbers of hosts(60,40,20 and 10). As we know that in Class C we have only the last octet to divide.

11111111. 11111111. 11111111.00000000

Subnets 1: (60 hosts)

Keeping process in mind of subnetting using VLSM stated in previous document we should borrow **two bits** to network bits so networks bit will become **26** as following

11111111. 11111111. 11111111.00000000

Subnet Mask: 192.168.3.0/26

32(total bits in ip address) - 26 (network bits) ⇒ 6 ⇒ 2^6=64 (hosts)

So first subnet will contain 64 IPs (62 usable IPs)

192.168.3.0 **Network IP**

192.168.3.1 . . . .192.168.3.62 **IP range for usable IPs(62 hosts)**

192.168.3.63 **Broadcast IP**

So we can allocate 60 IPs being in this subnet having 62 total usable IPs

Subnets 2: (40 hosts)

Similarly for 40 humber of hosts we can allocate another subnets having total 62 usable IPs as created above.

192.168.3.64 **Network IP**
192.168.3.65 . . . .192.168.3.126 **IP range for usable IPs(62 hosts)**
192.168.3.127 **Broadcast IP**

Subnet Mask: 192.168.3.64/26

Subnets 3: (20 hosts)

For 20 hosts we can borrow one more host bit to network bit as

11111111. 11111111. 11111111.00000000

32(total bits in ip address) - 27 (network bits) ⇒ 5 ⇒ 2^5=32 (hosts)

So the next network chunk will have 32 IPs which we can assign to 20 devices in the third subnet.

192.168.3.128 **Network IP**
192.168.3.129 . . . .192.168.3.158 **IP range for usable IPs(62 hosts)**
192.168.3.159 **Broadcast IP**

Subnet Mask: 192.168.3.128/27

Subnets 4: (10 hosts)

For this small subnet where we required only 10 usable IPs. We can borrow one more bit to network bits for smaller subnet.

11111111. 11111111. 11111111.00000000

32(total bits in ip address) - 28 (network bits) ⇒ 4 ⇒ 2^4=16 (hosts)

192.168.3.160 **Network IP**
192.168.3.161 . . . .192.168.3.174 **IP range for usable IPs(62 hosts)**
192.168.3.175 **Broadcast IP**

Subnet Mask: 192.168.3.160/28

NOTE: Using this VLSM technique we can minimize IP wastage. As in class C there are 256 total IPs we can allocate to different hosts.If we calculate total IPs used in this scenario which consumes **176** IPs (64+64+32+16). We have spare **80** (256-176) IPs which we can allocate to different hosts so we can create another subnet.

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